

Ghilmo Choi

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Education

Korea Aerospace University, Republic of Korea

Mar 2021 – Feb 2027

B.S. in Mechanical Engineering

(expected)

- Overall GPA: **4.08 / 4.5** (Major GPA: **4.18 / 4.5**)
- Main Coursework: Introduction to Machine Learning, Automatic Control, Mechanical Vibrations, Data Structures and Algorithms, Probabilistic Robotics

Research Interests

- Machine learning, deep learning, and AI model architectures for robotic intelligence
- Reinforcement learning for robotic decision-making and control
- Vision-and-language navigation and probabilistic localization
- On-device vision-language models for natural-language robotic mission execution
- Perception, Prediction, and Decision Making for Autonomous Driving

Research / Project Experience

MLCS Lab

Yonsei University

- *Undergraduate Intern*

July 2026 – Present

- **Model-based reinforcement learning under variable delay**
Investigating robust policy learning in environments with changing observation and action delays
- **Uncertainty-aware world-model rollouts**
Reproducing and adapting the MACURA framework with ensemble world models, model uncertainty estimation, uncertainty-aware synthetic rollouts, and SAC-based policy optimization
- **Adaptive model utilization**
Exploring delay-uncertainty estimation to adapt model trust, rollout length, and model-data ratio

ASOL (Autonomous Systems and Optimization Lab)

Korea Aerospace Univ.

- *Undergraduate Researcher*

June 2025 – July 2026

- **On-device VLM-based robotic mission execution**
Initiated at ASOL and continuing as a Capstone Design project, developing a natural-language mission execution framework that converts user commands into executable robot behaviors on a UGV platform
- **Hallucination-aware and SLAM-grounded task planning**
Designed an Actor–Guard–Critic planning loop and integrated RTAB-Map, object detection, structured world states, and ROS2/Nav2 control for real-robot language-to-motion execution
- **AirSim-based perception, reconstruction, and path generation**
Extending the framework by adopting open-vocabulary object detection models such as YOLOE, NanoOWL, and Grounding DINO on AirSim data, NVBlox for 3D reconstruction, and the OMPL library for path generation
- **Discrete Bayes filter-based localization for vision-and-language navigation**
Currently investigating probabilistic localization for VLN by maintaining and updating a discrete belief distribution over candidate locations, with plans to integrate the localization module into a navigation pipeline

KAUVOY Autonomous Driving Club

Korea Aerospace Univ.

- *Localization Part*

Dec 2024 – Nov 2025

- **EKF-based localization**
Implemented Dual EKF localization on an ERP42 autonomous vehicle by fusing IMU, wheel encoder, and RTK-GPS measurements
- **Autonomous parking and path tracking**
Developed and tested a real-vehicle parking pipeline covering cone perception, parking-space decision, Hybrid

A* path planning, and Pure Pursuit-based path tracking

KAUVOY Autonomous Driving Club

- *Team Leader*

Korea Aerospace Univ.

Dec 2025 – Present

- **MORAI-based autonomous driving development**

Primarily developing MPPI-based obstacle avoidance, LTV MPC-based path tracking, and FSM-based decision making in the MORAI simulator

Course Project, Introduction to Machine Learning

Korea Aerospace Univ.

- **End-to-End Machine Learning Classification and Regression Pipeline**

Aug 2025 – Nov 2025

- Built and validated classification and regression pipelines using preprocessing, dimensionality reduction, ensemble methods, and FCNNs; achieved **0.915 macro F1** on a 4-class imbalanced classification task

Honors & Activities

- Finalist, **2025 College Mobility Innovation Competition**
- Excellence Award, **Drone X AI Marketing Challenge** (2026)
- 2026 - **Airbus 101 Competition** (5th Place)

Publications

- **Co-author**, "*On-Device Vision Language Model Based Natural Language Mission Execution Framework*," submitted to the International Conference on Control, Automation and Information Sciences (**ICCAIS**), 2025
- **Co-author**, "*ROS robot_localization Dual EKF: Verification of Localization Accuracy and Application to Autonomous Driving Path Tracking*," submitted to the Korean Society of Automotive Engineers (**KSAE**), 2025

Technical Skills

- Python, C/C++, MATLAB, Linux(Ubuntu), ROS1, ROS2, Docker, Git